

Tony Peonio

Golden, CO

Tony_Peonio@Mines.edu, <https://github.com/TonyPeonio>

Skills

Languages: C, C++, Java, Python, Assembly, Bash, RosettaScripts

Skills: HPCs, Git/GitHub, LAMMPS, ASE, VESTA, Solidworks, Neural Networks, Pymol, PyTorch

Education

Colorado School of Mines 08/2024-05/2027

B.S. Computer Science (Quantum Engineering Minor) Golden, CO

- **GPA:** 4.00
- **Community Involvement:** Varsity Track & Field, Society of Quantum Engineers President, Quantum Engineering Student Advisory Board Member, Student-Athlete Advisory Committee, Bioinformatics Club Member, Mindful Eating And Living Club Treasurer
- **Grand Challenge Scholars Program:** Engineering Better Medicines focus

Relevant Experience

Rosetta Commons Research Experience for Undergraduates (NSF REU) 06/2026-PRESENT

Research Assistant Eugene, OR

- Presented a lightning talk and poster, "*Computational design of FPR2-specific cyclic peptides for resolution of chronic inflammation*," at RosettaCon 2026.
- Built multiple reproducible Python/Bash/RosettaScripts workflows for HPC clusters for large-scale peptide design and candidate prioritization.
- Invited to continue working remotely after the termination of the internship.

Colorado School of Mines & Kirin Autonomy 08/2026-PRESENT

CS Field Session Software Engineer Golden, CO

- Developing a chess neural network to dynamically assess player skill level and adapt AI difficulty in real-time.
- Developing rating-estimation algorithms and inference logic to analyze tactical decisions.

Colorado School of Mines Lincoln Carr Research Group 02/2026-PRESENT

Undergraduate Researcher Golden, CO

- Research involves SLM communication to search for emergent properties in interacting agentic AI communities, especially those relating to long-term internet manipulation.

Colorado School of Mines AMBER Research Group 09/2025-05/2026

Research Assistant Golden, CO

- Engineered secure, user-facing applications in Kotlin and Python to facilitate personal data access for air quality monitoring and research.

Colorado School of Mines Honors VIPER Quantum Materials 08/2025-05/2026

Student Researcher Golden, CO

- Worked in research focused on the computational discovery of novel quantum materials.

Awards and Honors

First Place Finish at Mines Protein Engineering 2026 Competition 01/2026-05/2026

- Secured first place in a university-wide protein engineering competition, engineering a *de novo* protein to bind to nitrogenase as the first step to cheaper nitrogen fixation.